

EcoPlate AI: Food Waste Intelligence Agent

Deliverable 1: Complete Lean Canvas (India Region - INR Currency)

1. Problem & Existing Alternatives

- High Food Waste in Indian Messes: Universities (IITs, NITs) & IT Parks discard 25-35% food daily.
- Lack of Itemized Disposal Telemetry: Wardens log total weight without tracking specific dishes.
- Financial & Environmental Toll: High procurement spend and rotting methane emissions violating FSSAI & SEBI BRSR.
- Existing Alternatives: Manual registers, basic weighing scales lacking dish vision classification.

2. Customer Segments & Early Adopters

- Target Customers: University Dining Messes, Corporate IT Cafeterias, Hospital Messes, Large Buffet Operators.
- Key Stakeholders: Mess Committee Leads, Catering Contractors (Sodexo India, Compass Group India), CSR Heads.
- Early Adopters: Top Higher Ed Campuses with Zero-Waste targets, Fortune 500 Corporate Hubs.

3. Unique Value Proposition (UVP)

- Zero-Touch Food Waste Intelligence: Autonomous multi-agent AI that sees what Indian kitchens waste, predicts what they need, and cuts procurement costs by up to 25-30% while advancing UN SDG 12 & FSSAI compliance.

4. Solution & Top Features

- Edge AI Vision Analysis: Overhead camera with fine-tuned YOLOv8 model for real-time dish recognition.
- Autonomous Predictive Procurement Agent: Multi-agent system analyzing historical waste, weather, and exam schedules.
- Automated Executive & Kitchen Reporting: FastAPI backend & React dashboard pushing portion alerts directly to mess chefs.

5. Channels & Revenue Streams

- Channels: Direct sales to Sodexo India/Compass Group India, IBM Ecosystem showcases, Higher Ed Summits.
- B2B SaaS Subscription: INR 25,000 - INR 60,000 / mess / month.
- Hardware & Setup Fee: INR 85,000 - INR 1,80,000 upfront per smart bin installation.

6. Cost Structure & Key Metrics

- Costs: Edge devices (Raspberry Pi 5/NVIDIA Jetson), Serverless cloud hosting, R&D for Indian regional datasets.
- Key Metrics: Kg Waste Prevented, Monthly Procurement Savings (INR 1.5L - 4.5L/mo), Vision Accuracy (>94%), CO2e Avoided.